Making Wage Rules Legible: Development, CPU-GPU Optimization, and Evaluation of a RAG Chatbot for Minimum-Wage Workers in Metro Manila

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***Abstract*—This study presents the development, optimization, and final evaluation of Making Wage Rules Legible, a retrieval-augmented generation chatbot designed to make Philippine wage rules and statutory benefits easier for minimum-wage workers in Metro Manila to access. The system used Qwen3-1.7B, a Chroma vector database, and three official labor documents covering the Labor Code of the Philippines, Wage Order NCR-27, and workers' statutory monetary benefits. Six retrieval configurations were examined by varying chunk size, overlap, and Top-k while retaining the same documents, questions, model, prompt, and generation settings. Each configuration was tested on Google Colab CPU and T4 GPU runtimes using a warm-up followed by three measured repetitions. Configuration C, with a chunk size of 1,200 characters, an overlap of 150, and a Top-k value of 3, offered the strongest balance between response quality and execution speed. Its average response time decreased from 467.5130 seconds on CPU to 8.1125 seconds on GPU, representing a 57.63-times speedup. After this configuration was selected, the completed chatbot was evaluated using ten worker-oriented questions, including eight answerable and two unanswerable questions. Seven responses were correct, one was correct with a citation issue, one was partially correct, and one was incorrect. The final evaluation recorded an average response time of 6.1019 seconds. The results were encouraging, particularly when the chatbot retrieved clear statutory provisions, but they also revealed weaknesses in numerical interpretation, citation precision, and the handling of unsupported additional statements. These observations suggest that efficient retrieval and GPU acceleration can make the system practical within free Colab resources, although stronger evidence validation remains necessary before it can be relied upon for actual worker guidance.**

Keywords*—Retrieval-augmented generation, RAG chatbot, CPU-GPU optimization, chunk size, chunk overlap, vector retrieval, Qwen3-1.7B, chatbot response evaluation, minimum-wage information, workers' statutory monetary benefits.

I. Introduction

Minimum-wage workers often need clear information about working hours, wage rates, holiday pay, and other statutory benefits, yet these rules are commonly presented in lengthy legal documents and technical tables. Although the information is publicly available, locating the correct provision and interpreting it in a practical situation can still be difficult. This concern inspired *Making Wage Rules Legible: A RAG Chatbot for Minimum-Wage Workers in Metro Manila*. The project explores how retrieval-augmented generation can make labor information easier to search by retrieving relevant passages from the Labor Code of the Philippines [1], Wage Order NCR-27 [3], and the Workers' Statutory Monetary Benefits Handbook [2] before generating an answer. Grounding the response in retrieved evidence is important because workers may rely on the information when understanding their wages and benefits.

Developing a useful RAG chatbot involves more than selecting a language model. Chunk size, overlap, and the number of retrieved passages can influence whether the system finds the correct legal provision or introduces unrelated information. Runtime performance also matters because a technically accurate system may still feel impractical when each response takes several minutes. This study therefore examined six retrieval configurations using the same documents, questions, Qwen3-1.7B model, prompt, and generation settings. Each configuration was evaluated on Google Colab CPU and T4 GPU runtimes, with attention given to indexing time, retrieval time, generation time, total response time, and answer quality. Through this comparison, the study aimed to identify a configuration that provides a thoughtful balance between speed, retrieval relevance, and source-supported answers rather than selecting a setting based on execution time alone. After the optimization experiment identified the most suitable configuration, the study continued by examining how the completed chatbot responded to a broader set of worker-oriented questions. Ten questions were prepared, including questions that could be answered from the indexed documents and questions for which the available evidence was intentionally insufficient. This additional

evaluation made it possible to observe not only whether the chatbot could retrieve and explain a relevant provision, but also whether it could recognize the limits of its knowledge base. Looking at both situations felt important because a labor-information assistant should be able to provide a supported answer when evidence is available and respond cautiously when it is not. The final evaluation therefore considered answer correctness, retrieval relevance, source support, and the handling of unanswerable questions alongside the earlier CPU and GPU performance measurements.

II. Model Background

The original model proposed for the project was gpt-oss-20b. However, the GPU resources available through Google Colab's free tier were not sufficient for the intended experiment. Since the activity required repeated testing across several retrieval configurations on both CPU and GPU, using a model that could run reliably within the available environment became an important practical consideration. Qwen3-1.7B was therefore selected as a smaller open-source alternative [4]. Its size allowed the complete RAG pipeline to run on a Colab CPU and T4 GPU while still supporting document-grounded question answering.

The language model works together with the retrieval components rather than answering from its internal knowledge alone. Source documents are divided into overlapping chunks and converted into numerical embeddings using a sentence-transformer model [5], after which the embeddings and source metadata are stored in a Chroma vector database [6]. When a worker submits a question, the same embedding model represents the query and retrieves the most similar passages. These passages are then included in the prompt given to Qwen3-1.7B. This arrangement helps direct the model toward the supplied labor documents, although it does not guarantee factual correctness. The experiment therefore considered both generated answers and retrieved evidence, particularly because OCR errors, unsuitable chunk boundaries, or irrelevant passages can still influence the model's interpretation. The CPU and GPU tests used the same model and generation settings so that observed performance differences could be associated mainly with the execution device and retrieval configuration. The same Qwen3-1.7B model was retained when the selected configuration was incorporated into the final chatbot interface. The interface allowed a worker to enter a question in Google Colab and view the generated answer, retrieved source documents, page references, and measured response times. Keeping the model and prompt consistent between the optimization and final-evaluation phases helped preserve a clear connection between the earlier experimental findings and the behavior of the completed chatbot. In this way, the final evaluation reflected the practical use of the configuration that performed most strongly during the controlled CPU and GPU comparison.

III. Experimental Methodology

A. Dataset Description

The experiment used three documents related to wage rules and statutory benefits in the Philippines: the Labor Code of the Philippines [1], Wage Order NCR-27 [3], and the Workers' Statutory Monetary Benefits Handbook 2024 Edition [2]. Together, these sources covered normal working hours, minimum-wage adjustments in Metro Manila, holiday pay, and other employee benefits. The Wage Order was image-based, so optical character recognition was applied before indexing to make its contents searchable. The same OCR-processed file and document-loading procedure were retained throughout all experiments. This consistency mattered because changing the document preparation between runs could influence retrieval results and make the configurations difficult to compare fairly.

B. Chunking Configurations

Six retrieval configurations were evaluated by changing chunk size, chunk overlap, and top-k. Configuration A used 400 characters, an overlap of 50, and top-k 3. Configuration B used 800 characters, an overlap of 100, and top-k 3, while Configuration C used 1,200 characters, an overlap of 150, and top-k 3. Configuration D retained a chunk size of 800 but increased the overlap to 200 to observe the effect of greater contextual repetition. Configuration E repeated the 800-character and 100-overlap setting but increased top-k from 3 to 5. A customized Configuration F was also included using a chunk size of 1,000, an overlap of 200, and top-k 3. All configurations used the same embedding model, Chroma vector database, Qwen3-1.7B model, source documents, and generation settings.

C. Evaluation Method

Three fixed questions representing easy, average, and difficult information needs were used for every configuration. These questions examined the normal eight-hour workday, the two wage rates under Wage Order NCR-27, and the calculation of regular-holiday pay using the first-tranche minimum wage. Each configuration was tested independently on Google Colab CPU and T4 GPU runtimes. One warm-up question was performed before measurement, after which every test question was executed three times. The experiment recorded indexing, retrieval, generation, and total response times, together with generated-token counts and retrieved passages. Answers were reviewed as correct, partially correct, or incorrect by comparing them with predetermined answers and supporting provisions. GPU speedup was calculated by dividing the CPU average time by the corresponding GPU average time. This evaluation considered both efficiency and answer quality, since a faster response was not treated as an improvement when its information was incorrect, incomplete, or unsupported.

D. Final Chatbot Evaluation Method

The final chatbot was evaluated using Configuration C, which applied a chunk size of 1,200 characters, an overlap of 150 characters, and a Top-k value of 3 on the Google Colab T4 GPU. Ten worker-oriented questions were prepared to examine the completed system beyond the three fixed questions used during the optimization experiment. Eight questions were answerable from the indexed documents, while two were intentionally unanswerable so that the chatbot's response to insufficient evidence could also be observed. The questions covered working hours, meal periods, weekly rest, service incentive leave, thirteenth-month pay, holiday pay, premium pay, and minimum-wage information. Each generated response was compared with a prepared reference answer, the expected source documents, and the passages retrieved by the system. The manual assessment considered answer correctness, retrieval relevance, source support, and the handling of unanswerable questions. Responses were classified as correct, correct with a citation issue, partially correct, or incorrect. This review was useful because a response could present the correct main fact while still including an inaccurate citation, an unsupported explanation, or an incorrect calculation.

IV. Results And Discussion

A. CPU and GPU Optimization Results

The experiments showed a clear performance difference between the CPU and T4 GPU runtimes. Across all six configurations, the GPU reduced average response time from several minutes to approximately eight to ten seconds. The amount of improvement varied because each configuration generated answers of different lengths and retrieved different amounts of context. Even with this variation, every configuration produced a GPU speedup greater than one, confirming that the T4 was consistently faster for the tested workload.

TABLE I. Average Response Time And GPU Speedup

Run	*CPU TIme (s)*	*GPU Time(s)*	*GPU Speedup*
A	314.5434	8.7679	35.87x
B	374.0495	8.1394	45.96x
C	467.513	8.1125	57.63x
D	384.8468	8.9699	42.90x
E	615.21	9.7487	63.11x
F	455.9458	8.3925	54.33x

Configuration E achieved the highest calculated response-time speedup at 63.11 times because its CPU execution was particularly slow. However, its GPU response time of 9.7487 seconds was also the slowest among the GPU runs. This observation shows why speedup should be interpreted together with the original execution times. Configuration C recorded the fastest GPU average response at 8.1125 seconds, followed closely by Configuration B at 8.1394 seconds. The small difference between them suggests that response speed alone was not enough to distinguish the more suitable configuration.

TABLE II. Indexing Time And GPU Speedup

Run	*CPU TIme (s)*	*GPU Time(s)*	*GPU Speedup*
A	152.2807	17.2371	8.83x
B	106.7095	15.4651	6.90x
C	90.5029	15.2925	5.92x
D	172.484	15.03	11.48x
E	116.9452	20.5445	5.69x
F	108.9922	15.8611	6.87x

GPU indexing was faster for every configuration, although its improvement was smaller than the response-generation improvement. Configuration D showed the largest indexing speedup at 11.48 times, while Configuration E showed the smallest at 5.69 times. Some variation was expected because document loading and text splitting still involve CPU-bound operations, while Colab resource allocation may also differ between sessions. The recorded measurements were retained as observed instead of being adjusted, since these variations reflect the practical conditions of using the free Colab environment.

TABLE III. Answer Quality Summary

Run	*CPU Answer Quality*	*GPU Answer Quality*
A	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
B	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
C	1 correct, 1 partial, 1 incorrect	2 correct, 1 partial
D	0 correct, 1 partial, 2 incorrect	0 correct, 1 partial, 2 incorrect
E	0 correct, 2 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
F	1 correct with a citation issue, 1 partial, 1 incorrect	2 correct with citation issues, 1 partial

Answer quality did not improve consistently as chunk size, overlap, or top-k increased. Configuration D showed

that increasing overlap from 100 to 200 at a chunk size of 800 did not improve retrieval and led to weaker answers. Configuration E also showed that retrieving five passages instead of three increased response time without resolving the main factual errors. Configuration F produced correct answers for two GPU questions, but its citations were less precise. Configuration C provided the strongest observed quality, with two correct GPU answers and one partially correct answer while also producing the fastest GPU response time.

The average-difficulty wage-order question remained partially correct or incorrect across the configurations. OCR interpreted portions of the scanned wage table incorrectly, causing the model to confuse the ₱60 wage increase with the resulting ₱755 daily wage. This result was particularly meaningful because it showed that retrieval optimization cannot fully correct inaccurate source text. Careful OCR validation and document preparation remain essential when a RAG system is used for legal or wage-related information.

B. Final Chatbot Evaluation Results

After Configuration C was selected, the completed chatbot was evaluated using ten worker-oriented questions on the T4 GPU. The system completed all ten questions, including eight answerable and two unanswerable cases. Seven responses were assessed as correct, one was correct with a citation issue, one was partially correct, and one was incorrect. The average retrieval time was 0.0122 seconds, while average generation time was 6.0849 seconds and average total response time was 6.1019 seconds. These measurements showed that the final chatbot remained responsive under the selected configuration. The answer-quality results were also encouraging because most questions were answered accurately and with relevant supporting passages. At the same time, the remaining errors showed that fast retrieval and generation did not remove the need to examine calculations, citations, and unsupported additions carefully.

TABLE IV. FINAL CHATBOT RESPONSE EVALUATION

Question ID	***Difficulty***	***Type***	***Final Assessment***
E01	Easy	Answerable	Correct
E02	Easy	Answerable	Correct with citation issue
E03	Easy	Answerable	Correct
E04	Easy	Answerable	Correct
E05	Average	Answerable	Correct
E06	Average	Answerable	Correct
E07	Average	Answerable	Correct
E08	Difficult	Answerable	Incorrect
E09	Difficult	Unanswerable	Correct
E10	Difficult	Unanswerable	Partially correct

The strongest responses were observed in E01 and E03 through E07, where the chatbot returned the expected information and the retrieved passages directly supported its conclusions. These questions covered normal working hours, weekly rest, service incentive leave, thirteenth-month pay, regular-holiday pay, and premium pay on a special non-working day that also fell on a rest day. E09 also showed a useful form of restraint. When asked for the current minimum wage in Cebu City, the chatbot recognized that the indexed sources did not provide the requested regional rate and avoided inventing an amount. E02 reached the correct conclusion that a regular meal period should generally be at least sixty minutes, but it referred to Article 8 instead of Article 85. This result was therefore classified as correct with a citation issue.

The more difficult cases revealed where further improvement is needed. In E08, the model correctly calculated 200% of ₱755 as ₱1,510 but then added the ₱755 basic wage again, producing an incorrect total of ₱2,265. The retrieved passages also failed to include Wage Order NCR-27, even though the question required evidence from both the wage order and the statutory-benefits handbook. In E10, the chatbot initially acknowledged that the indexed documents did not contain the employer's exact payroll dates, but it then introduced an unsupported statement that salaries are commonly released on the last day of the month. These responses suggest that the system can often identify the relevant rule or recognize missing evidence, yet it may still add an incorrect calculation or an unsupported explanation afterward. This observation reinforces the importance of reviewing the entire generated response rather than judging it only by its first correct statement.

V. RECOMMENDED CONFIGURATION

Based on the combined timing and answer-quality results, Configuration C using a chunk size of 1,200 characters, an overlap of 150 characters, and top-k of 3 is recommended together with the Google Colab T4 GPU runtime. This configuration produced the fastest measured GPU average response time of 8.1125 seconds and completed indexing in 15.2925 seconds. Its average response was approximately 57.63 times faster than its corresponding CPU result. More importantly, it provided the strongest observed answer quality, correctly answering the normal-workday and regular-holiday-pay questions while giving a partially correct response to the wage-order question.

The recommendation was not based on speed alone. Configuration E produced a larger calculated speedup, but retrieving five passages increased its GPU response time and did not improve the factual quality of its answers. Configuration F also performed well, although its generated citations were less precise. Configuration C appeared to provide enough context for the model without introducing as much unrelated or repetitive information. The T4 GPU is therefore recommended for practical use because it reduced response-generation time considerably while remaining available within the tested Colab environment. Nevertheless, the OCR-affected wage-order question should be treated carefully. Improving and validating the OCR text

is still necessary before the chatbot can be considered reliable for actual worker guidance.

The ten-question final evaluation provided an additional view of how Configuration C performed when used in the completed chatbot. Its average total response time of 6.1019 seconds remained practical on the T4 GPU, and most questions were answered correctly with relevant supporting passages. The configuration also allowed the chatbot to handle one of the unanswerable questions appropriately by recognizing that the requested Cebu City wage was outside the available evidence. However, the incorrect holiday-pay computation in E08 and the unsupported payroll statement in E10 showed that the selected configuration does not remove every generation error. Configuration C therefore remains the recommended setting because it provided the strongest overall balance observed in the experiments, but its outputs should still be treated as source-supported guidance that requires verification rather than as an automatic legal determination.

VI. Limitations And Possible Improvements

A. Limitations

The optimization phase used three fixed questions to maintain a controlled comparison across the six CPU and GPU configurations. The final chatbot evaluation expanded the test set to ten worker-oriented questions, including eight answerable and two unanswerable cases. Although this provided a broader view of the completed system, ten questions still represent only a small portion of the situations that workers may raise. The final questions were also executed once each, so the evaluation did not examine whether the same answers would remain consistent across repeated generations or different Colab sessions. Manual classification was necessary to examine correctness and source support, but this process may also involve some reviewer judgment. The most visible limitation came from the OCR-processed Wage Order NCR-27. Parts of its wage table were recognized incorrectly, causing the model to confuse the ₱60 wage increase with the resulting ₱755 daily minimum wage. Retrieval similarity also did not guarantee that the model interpreted a passage correctly. Some answers used the wrong pay rule, added an amount twice, or cited a passage that did not fully support the conclusion. A few responses reached the generation limit before presenting a final answer. In addition, CPU and GPU resources provided by the free Colab environment varied between sessions, which may have influenced model-loading, indexing, and response times. The experiment used repeated measurements and warm-up runs to reduce this effect, but the runtime conditions could not be controlled completely.

B. Possible Improvements

Future work should begin by validating the OCR output, particularly tables containing wage rates, dates, and monetary values. A cleaner text version or a table-aware OCR method could reduce the retrieval errors observed in the wage-order question. The evaluation set should also be expanded to include more worker-centered questions covering overtime, rest days, special non-working days, leave benefits, and wage deductions. English, Filipino, and Taglish questions could be included to better reflect how workers may communicate in practice. Retrieval may be improved through hybrid search, reranking, metadata filtering, and stronger source-selection rules. The prompt could also require the model to identify the applicable pay category before performing a calculation and to verify that every cited passage supports the final answer. Finally, future experiments could repeat each configuration across several Colab sessions and record one-time model-loading costs separately from normal question-answering time. These improvements would provide a broader and more reliable understanding of the system's practical performance.

The final response evaluation also suggested several focused improvements to the answer-generation process. For numerical questions, the prompt could require the model to identify whether a percentage represents additional compensation or the total pay rate before performing the calculation. It could then present the formula step by step and verify that the basic wage has not been counted twice, which would directly address the error observed in E08. For unanswerable questions, the model should be instructed to stop after explaining that the evidence is insufficient rather than adding a general practice or assumption, as occurred in E10. Citation validation could also compare the article number, source filename, and retrieved page before the answer is displayed, helping prevent citation mistakes such as the Article 8 reference in E02. Finally, reranking or source-aware retrieval could help ensure that questions requiring evidence from more than one document retrieve all necessary sources instead of relying on information already stated in the question.

VII. Conclusion

This study examined how chunk size, overlap, top-k, and execution device affected the performance of a RAG chatbot for Metro Manila wage and statutory-benefit information. Six configurations were tested using the same source documents, questions, Qwen3-1.7B model, and generation settings. The results showed that the T4 GPU substantially reduced indexing and response-generation times across all configurations. However, the experiments also demonstrated that faster execution did not automatically produce more reliable answers. Retrieval quality, OCR accuracy, and the model's interpretation of the retrieved passages remained important factors.

Configuration C, with a chunk size of 1,200 characters, an overlap of 150, and a Top-k value of 3, provided the strongest observed balance among the tested settings. On the T4 GPU, it achieved an average experimental response time of 8.1125 seconds and a 57.63-times speedup over its corresponding CPU result. This configuration was therefore carried forward into the completed chatbot, where it supported a Colab-based interface that displayed generated

answers, retrieved sources, page references, and response-time measurements.

The ten-question final evaluation provided a broader view of the chatbot's practical behavior. Seven responses were correct, one was correct with a citation issue, one was partially correct, and one was incorrect, while the average total response time was 6.1019 seconds. The chatbot performed well when the retrieved passages stated the required rule clearly and also handled one unanswerable question appropriately. However, the incorrect holiday-pay computation and unsupported payroll statement showed that relevant retrieval does not always guarantee a fully reliable answer. Overall, the project demonstrated that a smaller open-source model can support a responsive labor-information RAG chatbot within free Google Colab resources. The system remains an educational prototype, and further work on numerical verification, citation checking, OCR quality, retrieval across multiple documents, and cautious handling of insufficient evidence is necessary before it can be relied upon for actual worker guidance.

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